VRS-110: Controller firmware 5.1.0
Release notes and update instruction

Bulletin number	**SB-2026-003**
Category	**Reliability**
Issue date	**2026-01-27**
Product	**VRS-110 rotary screw compressor with VSD, Vcon 4 controller**
Firmware	**5.1.0 (build 5.1.0.2611), replaces 5.0.3 and earlier**
Action	**Update at next scheduled service visit (see 5)**
Revision	**0**

1 SUMMARY

Firmware 5.1.0 is a maintenance release for the Vcon 4 controller of the VRS-110. It contains corrections and a number of smaller improvements, which are listed in section 4. Vastra recommends that all VRS-110 units are updated at the next scheduled service visit. A dedicated visit is not necessary, with the exception of units described in 5.2.

Units delivered from February 2026 leave the factory with 5.1.0 installed. Release notes for the Vcon 3 controller of the VRS-45 and VRS-75 are published separately and are not part of this bulletin.

2 AFFECTED UNITS

Model	Serial number	Controller	Installed firmware	Recommendation
VRS-110	VRS110-210001 and later	Vcon 4	4.x	Update to 5.1.0. Read 3 first
VRS-110	VRS110-210001 and later	Vcon 4	5.0.0 to 5.0.3	Update to 5.1.0 at next service
VRS-110	VRS110-210001 and later	Vcon 4	5.1.0	No action

The installed firmware version is shown in Menu > Info > Software on the controller display, or in VastraLink under Unit > Properties.

3 COMPATIBILITY

- VastraLink service software 3.4 or later is required. Older versions can not open the 5.1.0 package and show alarm codes introduced in 5.1.0 only as "unknown alarm".
- Units on 4.x firmware must first be updated to 5.0.3 and then to 5.1.0. A direct update from 4.x is blocked by the boot loader.
- Parameter sets are kept at the update. Display language and service interval settings are reset to factory values and must be set again (see 4, items VC4-1902 and VC4-1925).
- Building management systems which read the active alarm register by Modbus will receive alarm number 15, which did not exist before. Inform the site so that the alarm list in the BMS can be extended.

4 RELEASE NOTES 5.1.0

Changes compared to 5.0.3. Items are listed in order of the internal reference number.

Ref.	Area	Type	Description
VC4-1873	Communication	Fixed	Modbus RTU: holding register 40112 (motor speed, rpm) returned wrong values above 3,276 rpm because of a signed integer overflow. Register is now unsigned.
VC4-1902	Display	New	Polish and Dutch added as display languages (Menu > Settings > Language). Czech is planned for a later release.
VC4-1911	Hour meter	Fixed	Running hours and loaded hours counters rolled over to 0 after 65,535 h. Counters are now 32-bit. Existing values are transferred at the update.

Ref.	Area	Type	Description
VC4-1925	Service	Changed	Service interval reminder can be set per oil type (Menu > Service > Intervals). Intervals for oil, oil filter, air filter and separator element are set separately. Use the values in the service manual of the model.
VC4-1940	Temperature supervision	Changed	TS-2 (discharge temperature) plausibility check added: signal range, open circuit, short circuit and rate of change. A TS-2 signal out of range now gives new alarm E-15 "TS-2 sensor fault" (compressor stops, auto restart blocked). E-14 "Discharge temperature high" is from now raised only for a confirmed over-temperature above 110 C with a plausible TS-2 signal. Note: in firmware 5.0.3 and earlier, a failing or drifting TS-2 reads high and is reported as E-14.
VC4-1952	Event log	Changed	Each alarm entry stores ambient temperature, discharge pressure and motor speed at the time of the alarm. USB export is now CSV with ISO 8601 time stamps.
VC4-1958	Fan drive	Fixed	After a supply interruption shorter than 0.5 s the cooling fan did not always restart, while the main drive restarted normally. Fan restart is now supervised.
VC4-1961	Real-time clock	Fixed	Controller clock lost up to 4 minutes per month. Daylight saving change for EU and North American time zones is now automatic (Menu > Settings > Time zone).
VC4-1966	Remote control	Changed	Modbus remote start/stop: watchdog timeout is configurable from 5 to 120 s (default 30 s, before fixed at 10 s). Reduces unwanted stops on slow site networks.
VC4-1970	Display	Fixed	Load and unload pressure were displayed rounded to 0.5 bar on the pressure band screen. Now shown with 0.1 bar.
VC4-1977	Security	Changed	Service level password minimum 6 digits. The factory default password must be changed at first login with VastraLink.
VC4-1981	Pressure control	Fixed	Compressor did not leave sleep mode if network pressure dropped during the first 10 s of sleep. Corrected.

5 ACTION

5.1 All units

Update to 5.1.0 at the next scheduled service visit, following the instruction in 6. Record the old and new firmware version in the service report.

5.2 Units with repeated E-14 on firmware 5.0.3 or earlier

On these firmware versions it is not possible to tell from the alarm code alone whether E-14 is caused by a real high discharge temperature or by a faulty TS-2 sensor. If a unit has shown E-14 more than one time, it is recommended to:

1. Check TS-2 (part 41-2207) against a calibrated contact thermometer at the discharge flange, as described in the VRS-110 Service Manual under troubleshooting of E-14. Replace TS-2 if it is outside the tolerance given there.
2. Check the normal causes of high discharge temperature in the same section (cooler fouling, oil level, thermostatic valve, ambient and ventilation).
3. Update the firmware to 5.1.0 at the same visit. Do not wait for the next scheduled service.
4. If E-14 or E-15 returns after the update, export the event log (VC4-1952) and send it to Vastra Technical Support with the serial number. Replacing TS-2 a further time without this information is not recommended.

6 UPDATE INSTRUCTION

Needed: Windows laptop with VastraLink 3.4 or later and a valid service login; firmware package VCON4_5.1.0.2611.vfw from the Vastra partner portal (check that the CRC32 shown in VastraLink is 7E1A90C4); the X10 service port cable (USB-A to M12, 8-pole) from the Vastra service tool kit. A standard USB cable can not be used.

1. Inform the site. The compressor is not available for approximately 30 minutes.
2. Stop the compressor with the normal stop button and wait until it is unloaded and stopped. Do not use the emergency stop.

3. Connect the X10 cable to service port X10 on the controller board, in the control compartment behind the left door. The VSD compartment does not need to be opened for the update.
4. In VastraLink, save the parameter set (Parameters > Save to file) and export the event log. Write down the running hours.
5. Start the update in VastraLink (Firmware > Update). The transfer takes approximately 12 minutes. The controller restarts two times. Do not disconnect the cable or switch off the supply during the update. If the update is interrupted, the controller starts in boot mode; repeat the update from step 5.
6. After the restart, check that Menu > Info > Software shows 5.1.0.2611. Compare the parameters with the saved file (Parameters > Compare). Set display language and service intervals again.
7. Start the compressor and run it loaded for at least 15 minutes. Check that no alarm is active.
8. Disconnect the cable, close the door and enter the update in the service report.

7 LABOUR ALLOWANCE AND WARRANTY

Case	Labour allowance	Claim code
Update during a scheduled service visit	0.5 h	VK-26003-FW
Update during a separate visit for a unit under 5.2	1.0 h + travel per agreement	VK-26003-FW
Update outside the compressor warranty period	Charged per customer agreement	-

The firmware is free of charge. Claims are made in the Vastra partner portal within 60 days, with old and new firmware version and the serial number. TS-2 sensors replaced under 5.2 are claimed under normal warranty if the unit is in warranty.

8 DISTRIBUTION

- Vastra authorised service partners: Nordic countries, Europe, North America
- Vastra Technical Support, Level 1 and Level 2
- Vastra sales companies and distributors
- Vastra Product Management Controls

9 REVISIONS

Rev.	Date	Description
0	2026-01-27	First issue

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Questions about this bulletin: Vastra Technical Support Level 2, Anders Holm, anders.holm@vastra.se, +46 21 480 22 10. Please have the serial number and the exported event log ready.